

Analysis of Low-Rank Structure in Pre-trained Transformer Projection Weights

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Abstract

This study investigates whether pre-trained Transformer projection weights exhibit low-rank structure, a property distinct from the empirically-validated success of low-rank adaptation methods such as LoRA. Through singular value decomposition analysis of Query, Key, and Value projection matrices in Mistral-7B (a 7B-parameter, 32-layer model architecturally similar to LLaMA), we measured effective ranks at 99% variance threshold for layers 20-31. Results show effective ranks ranging from 1554 to 1647, representing approximately 38-40% of the model dimension (4096). These values exceed the hypothesized threshold of $r_{\text{eff}} < 256$ by a factor of 6-7 $\times$. Additionally, operator entropy does not decrease monotonically with layer depth ($\beta = +0.001453$, $p = 0.072$). The findings indicate that pre-trained projection weights at 7B scale maintain near full-rank structure, contradicting assumptions underlying certain post-hoc architectural conversion techniques. This measurement clarifies that LoRA’s effectiveness with ranks 8-64 reflects properties of task-specific updates rather than compressibility of pre-trained weights themselves.

1. Introduction

Low-Rank Adaptation (LoRA) enables fine-tuning of large language models using rank-8 to rank-64 matrix decompositions, reducing trainable parameters by orders of magnitude while maintaining performance comparable to full fine-tuning. This empirical success has led to an implicit assumption in portions of the model

compression literature: if low-rank updates ΔW work effectively, perhaps pre-trained weights W themselves exhibit low-rank structure.

We tested this assumption directly through singular value decomposition analysis of projection weight matrices in Mistral-7B, a 32-layer decoder-only Transformer with 7B parameters. The analysis targeted deep layers ($L \geq 20$) where semantic compression theories predict the strongest low-rank signatures. Results show effective ranks of 1554-1647 at 99% variance threshold, approaching 40% of the model dimension (4096). This represents 6-7 $\times$ higher than thresholds commonly assumed for certain post-hoc compression approaches and 50 $\times$ higher than typical LoRA adaptation ranks.

1.0.1. SCOPE AND OBJECTIVES

This work measures a specific structural property: the effective rank of learned projection weight matrices in pre-trained models. The measurement addresses whether assumptions derived from LoRA’s success—namely, that pre-trained weights might be inherently low-rank—hold empirically at the 7B parameter scale.

We do not propose new methods, compare competing techniques, or evaluate model performance. This is a measurement study establishing empirical ground truth for a structural property relevant to compression research.

1.0.2. CONTRIBUTIONS

1. Direct measurement of effective rank in pre-trained 7B-scale Transformer projection weights, revealing near full-rank structure ($r_{\text{eff}} \approx 1600$)
2. Empirical refutation of the assumption that projection weight matrices exhibit low-rank structure at this scale
3. Validated SVD-based analysis methodology applicable to other architectural families and scales
4. Clarification that LoRA’s success with low-rank updates ($r = 8-64$) is independent from the rank

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structure of pre-trained weights ($r_{\text{eff}} = 1600$)

2. Related Work

2.0.1. PARAMETER-EFFICIENT FINE-TUNING

LoRA freezes pre-trained weights W and learns low-rank updates $\Delta W = BA$ where $B \in \mathbb{R}^{(d \times r)}$, $A \in \mathbb{R}^{(r \times d)}$, with r typically 8-64. The method demonstrates that task-specific adaptations can be captured in low-dimensional subspaces. LoRA does not explicitly claim that pre-trained weights are low-rank; it shows that updates can be low-rank. Our work measures the property that LoRA does not claim: the rank of W itself.

Subsequent methods including AdaLoRA and DyLoRA explore adaptive rank allocation, indicating that optimal rank varies by task and layer. The existence of rank-search methods suggests low-rank sufficiency is task-dependent rather than a fixed property of weights.

2.0.2. STATE SPACE MODELS AND HYBRID ARCHITECTURES

Mamba introduced selective state space models achieving linear-time complexity $O(L)$ with performance matching Transformers at certain scales. Hybrid architectures such as Samba interleave SSM blocks with attention layers, demonstrating complementary strengths when co-trained from initialization.

Post-hoc conversion of pre-trained Transformers to SSMs remains challenging. Our measurement provides one explanation: pre-trained Transformer weights do not exhibit the bounded-state low-rank structure ($r_{\text{eff}} < 256$) that certain SSM conversion approaches assume.

2.0.3. POST-HOC COMPRESSION

Magnitude pruning, structured pruning, and quantization methods often assume redundancy in pre-trained weights. Knowledge distillation achieves behavioral equivalence without requiring structural similarity. Our finding that projection weights maintain high effective rank ($r_{\text{eff}} = 1600$) does not preclude these methods but does inform approaches that explicitly assume low-rank weight structure.

3. Methodology

3.0.1. EFFECTIVE RANK COMPUTATION

For a weight matrix $W \in \mathbb{R}^{(d_{\text{out}} \times d_{\text{in}})}$, the singular value decomposition is $W = U\Sigma V^T$. The effective rank

at variance threshold τ is defined as:

$$\text{rank}(W, \tau) = \min r : \sum_{i=1}^r \sigma_i^2 / \sum_{i=1}^{\text{rank}(W)} \sigma_i^2 \geq \tau$$

We use $\tau = 0.99$ (99% variance threshold). This quantifies how many singular values encode 99% of the matrix information. We chose $r_{\text{eff}} < 256$ as the test threshold based on state size constraints in selective SSM architectures ($N \leq 1024$) and assumptions in post-hoc conversion literature.

3.0.2. TARGET MATRICES AND LAYERS

We analyzed Query (Q), Key (K), and Value (V) projection weight matrices in multi-head attention layers. For Mistral-7B with hidden dimension 4096 and 32 attention heads, each projection is $W_{Q,K,V} \in \mathbb{R}^{(4096 \times 4096)}$. These are learned parameters stored in the model, not runtime-computed attention patterns.

Analysis focused on layers 20-31 (deep layers where $L \geq 20$ in the 32-layer model). Semantic compression theories predict deep layers should exhibit the strongest low-rank signatures.

3.0.3. OPERATOR ENTROPY MEASUREMENT

We measured operator entropy via log-determinant of covariance matrices:

$$H(W) = \log \det((1/n) W^T W)$$

The hypothesis predicted monotonically decreasing entropy with layer depth. We tested this via linear regression $H(L) = \beta \times L + \text{intercept}$ with null hypothesis $H_0: \beta \geq 0$ and alternative $H_A: \beta < 0$, $p < 0.01$.

3.0.4. MODEL AND COMPUTATIONAL SETUP

Analysis used Mistral-7B-v0.1, a 32-layer decoder-only Transformer with 7B parameters, 4096 hidden dimension, and 32 attention heads. Mistral shares architectural characteristics with the LLaMA family (32 layers, similar hidden sizes).

SVD computation for 4096×4096 matrices used NumPy's `np.linalg.svd` with float64 precision for numerical accuracy. Analysis was conducted on NVIDIA H100 GPU for weight loading; SVD computation is CPU-bound.

3.0.5. TWO-PHASE VALIDATION

To ensure measurement reliability separates from hypothesis outcomes:

Phase 1 (h-e1): Methodology Validation

- Implement SVD-based rank computation pipeline